

AI-Based Waste Management Monitoring System

1. Project Overview

Project Title

Waste Management System

Domain

Artificial Intelligence / Computer Vision / Smart City Monitoring

Objective

The objective of this project is to detect illegal waste dumping activities in real time using CCTV streams and AI-based object detection systems. The system automatically identifies suspicious dumping behavior, records evidence clips, uploads them to cloud storage, and sends alerts to municipality authorities.

2. Problem Statement

Across municipalities and urban areas, people often dump waste illegally in unauthorized and undesignated locations. Manual monitoring using security personnel is inefficient, expensive, and not scalable.

Challenges include:

- Lack of continuous monitoring
- Delay in identifying dumping activities
- Difficulty in collecting evidence
- Manual reporting process
- High operational cost
- Increased environmental pollution

The project aims to solve these problems using AI-powered automated monitoring.

3. Solution Overview

The system uses RTSP CCTV streams as input and continuously monitors suspicious activities.

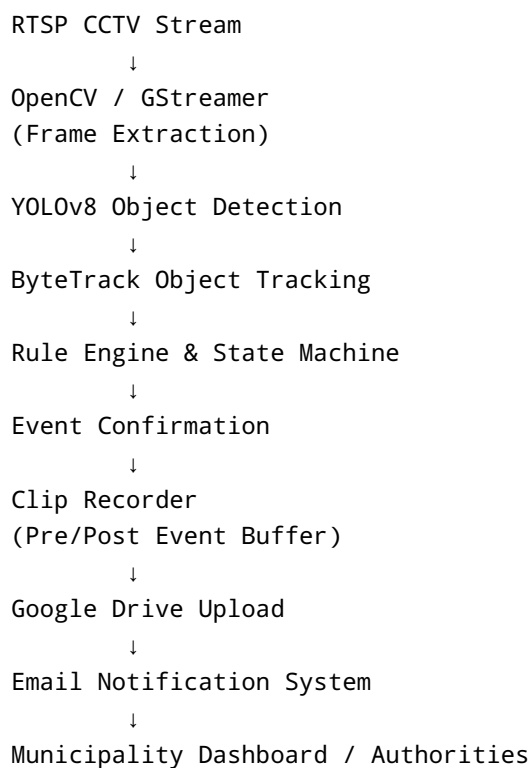
The workflow includes:

1. Capturing live CCTV footage

2. Detecting persons and waste-related objects using YOLOv8
3. Tracking objects using ByteTrack
4. Applying behavioral analysis logic
5. Confirming dumping events using a state machine
6. Recording pre-event and post-event clips
7. Uploading clips to Google Drive
8. Sending automated email notifications with evidence links

The system reduces false positives by validating activities across multiple frames instead of relying on single-frame detection.

4. System Architecture



5. Technology Stack

Category	Technologies Used
Programming Language	Python
Computer Vision	OpenCV
AI Model	YOLOv8
Object Tracking	ByteTrack

Category	Technologies Used
Streaming	RTSP CCTV Streams
Backend API	FastAPI
Notifications	Email Alerts / Firebase
Cloud Storage	Google Drive API
Configuration	Python dotenv (.env)
Video Processing	FFmpeg / OpenCV
Event Buffering	Deque

6. Core Modules

6.1 RTSP Stream Handler

Purpose

Handles live CCTV video streams.

Responsibilities

- Connect to RTSP stream
- Reconnect during stream failures
- Read video frames continuously

Technologies

- OpenCV
 - GStreamer
-

6.2 YOLOv8 Detection Engine

Purpose

Detects:

- Person
- Waste objects
- Vehicles
- Suspicious objects

Features

- Real-time detection

- High-speed inference
- Confidence threshold filtering

Output

- Bounding boxes
 - Class labels
 - Confidence scores
-

6.3 ByteTrack Tracking Module

Purpose

Tracks detected objects across multiple frames.

Why Tracking is Required

YOLO alone cannot confirm dumping behavior. Tracking helps:

- Maintain object identity
 - Monitor movement
 - Analyze behavioral patterns
 - Reduce false positives
-

6.4 Rule Engine

Purpose

Analyzes object behavior.

Conditions Used

- Person near object
- Object separation from person
- Object remains stationary
- Person walks away
- Object still exists after person leaves

Only after validating all conditions does the system confirm dumping.

6.5 State Machine

Purpose

Manages event lifecycle.

States

```
IDLE
TRACKING
OBJECT_ASSOCIATED
SUSPECTED_DUMP
CONFIRMED_EVENT
```

Benefits

- Reduces false positives
 - Prevents duplicate alerts
 - Improves event accuracy
-

6.6 Clip Recorder

Purpose

Stores evidence clips.

Features

- Pre-event recording buffer
- Post-event recording
- Efficient storage using deque
- Automatic MP4 generation

Configuration

- Pre-event buffer: 15 seconds
 - Post-event recording: 15 seconds
-

6.7 Google Drive Integration

Purpose

Uploads event clips automatically.

Workflow

1. Upload clip to Drive
2. Generate public link
3. Send link through email alert

Benefits

- Reduces email attachment size

- Centralized storage
- Easy accessibility

6.8 Email Alert System

Purpose

Sends automated notifications.

Includes

- Camera ID
- Location
- Event timestamp
- Google Drive clip link

Example Alert

Illegal dumping activity detected.

Location: Municipality Zone A

Camera ID: CAM-01

Evidence Link:

<https://drive.google.com/file/d/xxxxx/view>

7. Event Confirmation Logic

Workflow

```
Person detected
  ↓
Waste object detected
  ↓
Object close to person
  ↓
Object separated from person
  ↓
Object becomes stationary
  ↓
Person walks away
  ↓
Object still remains
```

↓
CONFIRMED DUMP EVENT

8. False Positive Reduction Strategy

The system avoids triggering alerts using:

- Multi-frame validation
- Object tracking
- State machine validation
- Confidence threshold filtering
- Stationary object analysis
- Time persistence checks

Example: If a person temporarily places a bag and picks it back up, the alert is cancelled.

9. Configuration Management

Environment Variables (.env)

Sensitive configurations are stored securely.

Example

```
GDRIVE_CREDENTIALS_FILE=credentials.json
GDRIVE_FOLDER_ID=folder_id

ALERT_SENDER_EMAIL=yourmail@gmail.com
ALERT_APP_PASSWORD=app_password

CAMERA_ID=CAM-01
LOCATION_NAME=Municipality Zone A
```

Benefits

- Security
- Easy deployment
- Environment separation
- No hardcoded secrets

10. Folder Structure

```
project/
├── app/
│   ├── detector.py
│   ├── tracker.py
│   ├── notification_manager.py
│   ├── drive_uploader.py
│   ├── email_alert.py
│   ├── clip_recorder.py
│   └── config.py
├── clips/
├── credentials.json
├── .env
├── main.py
└── requirements.txt
```

11. Security Considerations

- Credentials stored in .env
- Google Drive API authentication
- Controlled access to cloud storage
- Restricted municipality dashboard access
- Public area CCTV monitoring only

12. Current POC Features

Implemented Features:

- Real-time RTSP monitoring
 - YOLOv8 object detection
 - ByteTrack tracking
 - State machine event confirmation
 - Video clip recording
 - Google Drive upload
 - Email notification system
 - ROI-based monitoring
 - Configurable thresholds
-

13. Future Enhancements

Planned Improvements

- Kafka-based event streaming
 - Firebase Cloud Messaging (FCM)
 - Web dashboard
 - Multi-camera analytics
 - Cloud deployment
 - GPU acceleration
 - Advanced analytics dashboard
 - Mobile application integration
 - Real-time municipality monitoring panel
-

14. Key Advantages

- Fully automated monitoring
 - Real-time event detection
 - Reduced manual intervention
 - Scalable architecture
 - Reduced false alerts
 - Evidence-based reporting
 - Cloud-based accessibility
 - Smart city integration capability
-

15. Challenges Faced

Challenge	Solution
False positives	State machine + tracking
Video storage	Pre/post clip buffering
Large email attachments	Google Drive links
Stream disconnections	Auto reconnect logic
Object identity loss	ByteTrack integration

16. Performance Metrics

The model performance was evaluated using:

- Accuracy
- Precision

- Recall
 - F1 Score
 - mAP (Mean Average Precision)
-

17. Business Impact

This system can help municipalities:

- Improve cleanliness monitoring
 - Reduce illegal dumping
 - Improve operational efficiency
 - Enable faster response actions
 - Reduce manpower dependency
 - Support smart city initiatives
-

18. Demo Flow

End-to-End Demonstration

```
graph TD; A[RTSP CCTV Stream] --> B[Person dumps waste]; B --> C[YOLOv8 detects activity]; C --> D[ByteTrack tracks object]; D --> E[Rule engine confirms event]; E --> F[Clip recorded]; F --> G[Clip uploaded to Google Drive]; G --> H[Email notification sent]; H --> I[Authority reviews evidence];
```

19. Conclusion

The AI-Based Waste Management Monitoring System successfully demonstrates how artificial intelligence and computer vision can automate illegal waste dumping detection in real time.

By integrating YOLOv8, ByteTrack, behavioral analysis, cloud storage, and automated notifications, the system provides an efficient and scalable solution for smart municipality monitoring.

The project showcases strong practical implementation of:

- Real-time AI systems
- Computer vision pipelines
- Event-driven architecture
- Cloud integration
- Automated alert workflows

This solution has the potential to be expanded into a large-scale smart city monitoring platform.